

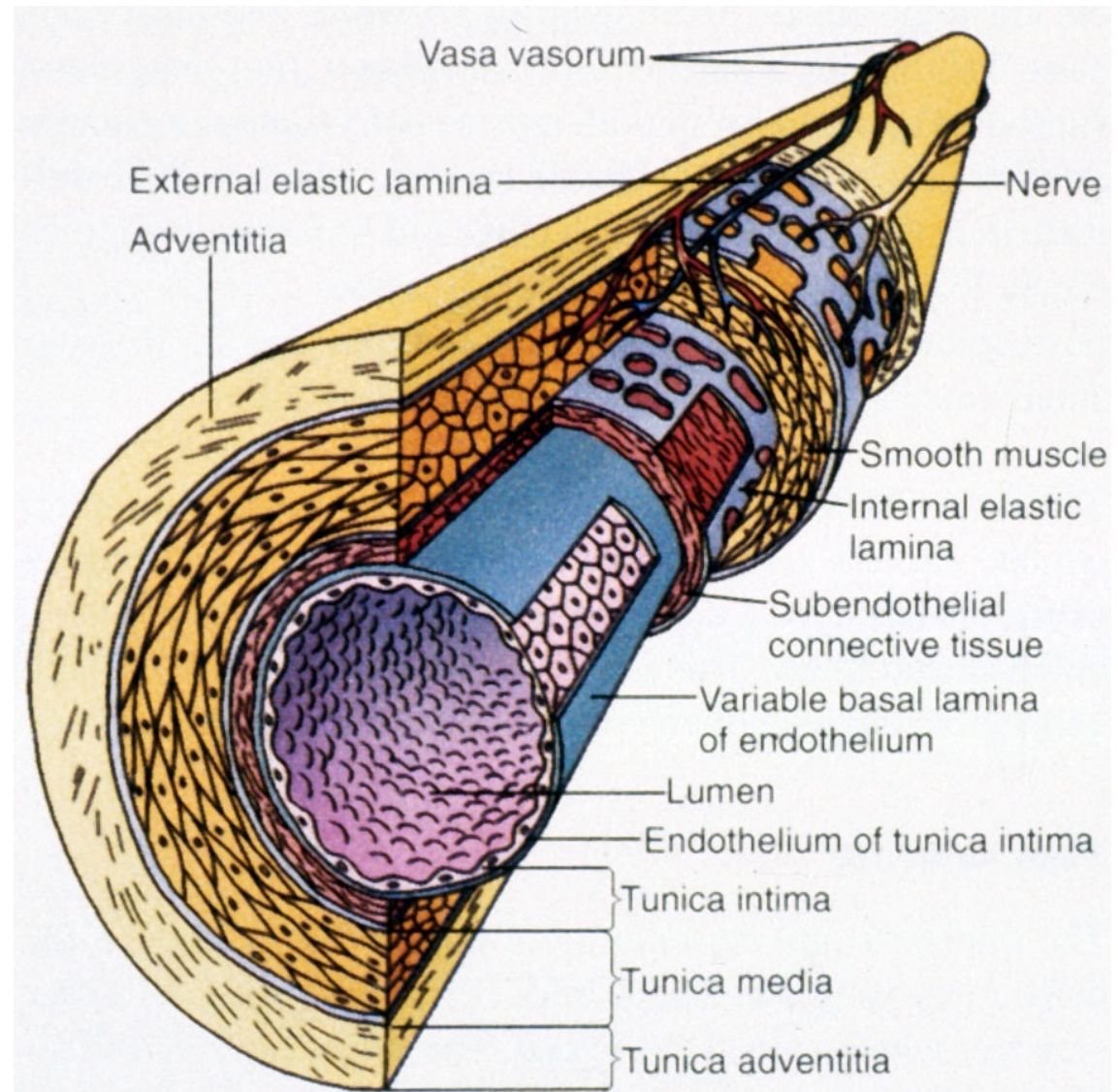
Histology of Blood and Lymphatic Vessels



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Blood Vessel Structure

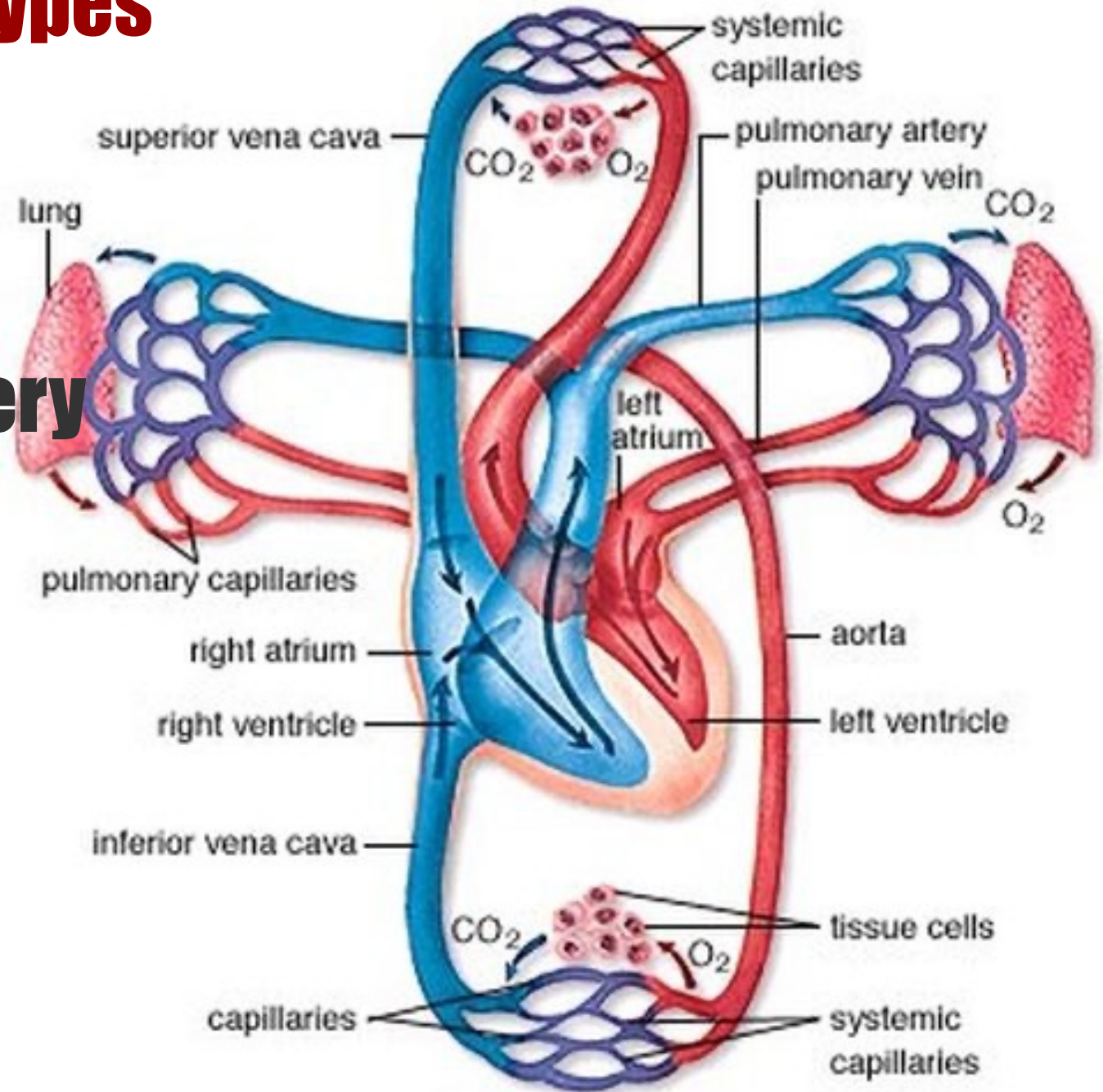
- Tunica Intima
 - Tunica Media
 - Tunica adventitia
-
- Vessels of different types vary in the presence of or thickness of these 3 layers.



Blood Vessel Types

Heart

- Elastic Artery
- Muscular Artery
- Arteriole
- Capillary
- Venule
- Medium Vein
- Large Vein
- Heart



Elastic Artery — these vessels are designed to withstand the surge of systolic contraction and absorb the full output volume of the ventricle.

Tunica Intima

Endothelium with basal lamina, sub-endothelial layer, incomplete internal elastic lamina

Tunica Media

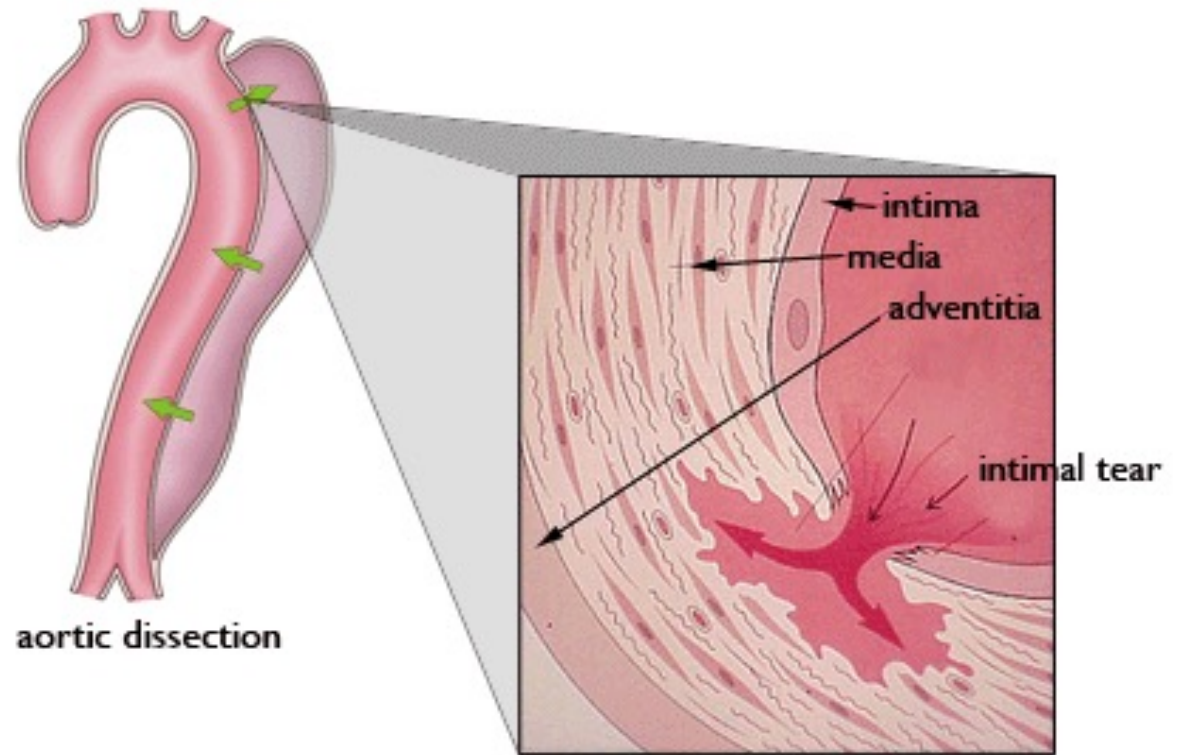
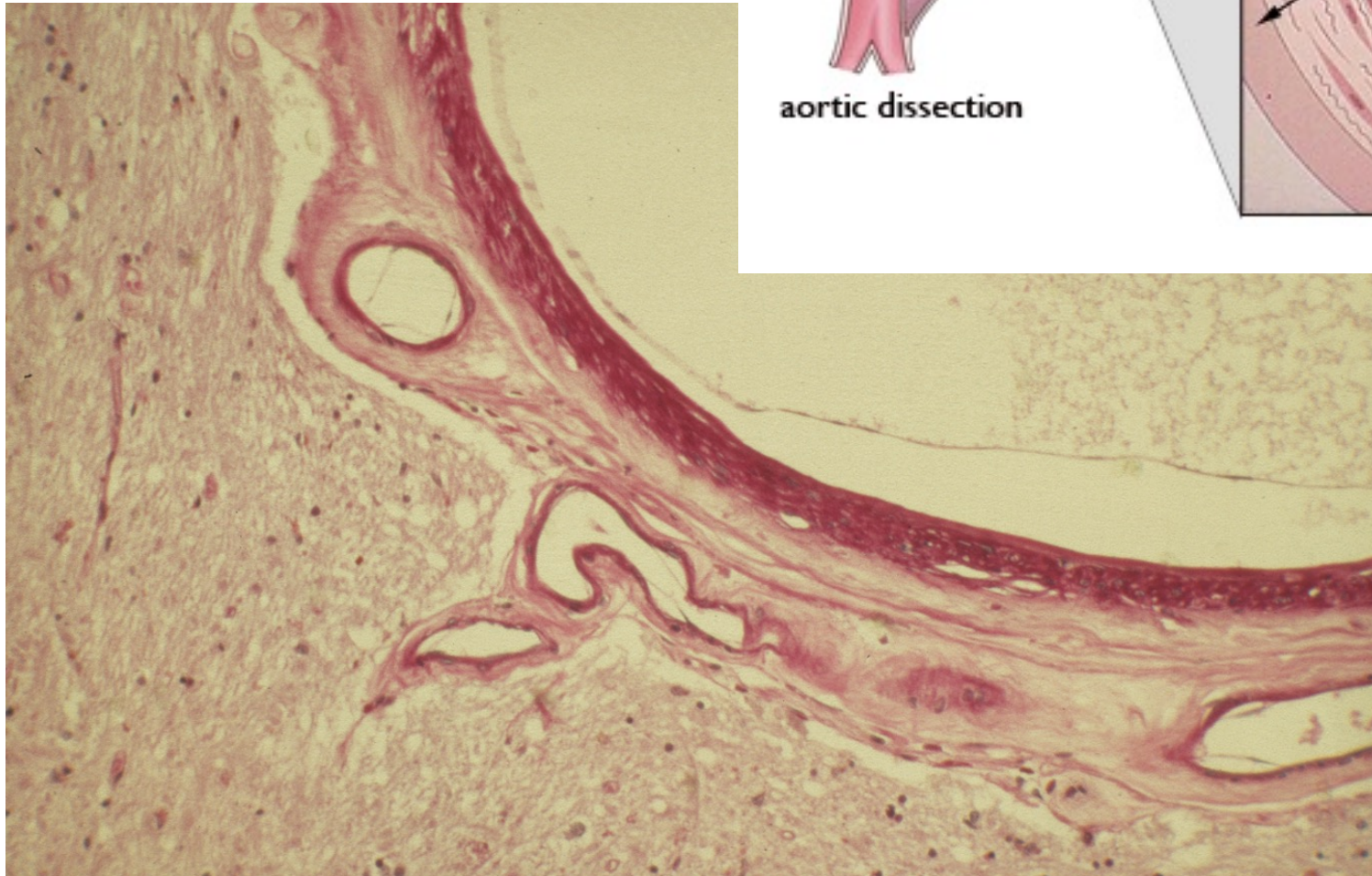
40-70 fenestrated elastic membranes, smooth muscle cells. Thin external elastic lamina, vasa vasorum

Tunica Adventitia

Thin layer of fibroelastic connective tissue, vasa vasorum

Vasa Vasorum

— The walls of vessels like the Aorta are thick enough that the blood in the lumen is insufficient to keep the thick wall alive. As such small vessels leave the aorta to pass within the wall of the aorta creating what are called vasa vasorum.

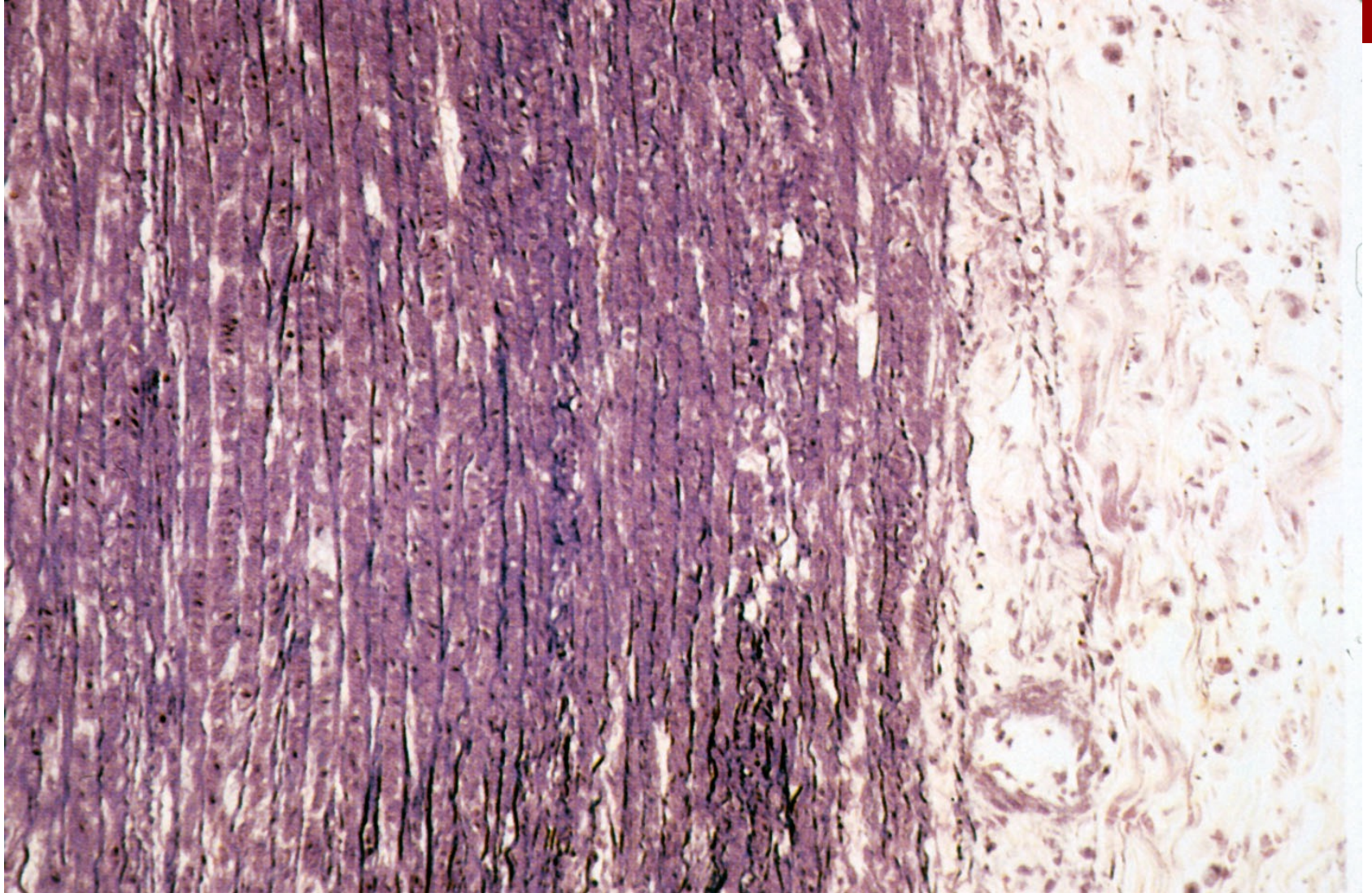


With age the openings of these vessels into the lumen of the aorta can tear, leading to a “dissecting” aneurysm.

Elastic Artery – aorta, proximal ends of carotids, vertebrals and brachiocephalic and subclavian arteries (vessels that feel the surge of systole)

Black filaments of elastin interweave throughout the tunica media

Lumen



Muscular Artery — most of the vessels large enough to see grossly and be named (coronary vessels, superficial temporal artery, radial, celiac, renal, maxillary, mesenterics, etc). These feel the pulse but do not have to withstand the initial surge.

Tunica Intima

Endothelium with basal lamina, subendothelial layer, THICK internal elastic lamina

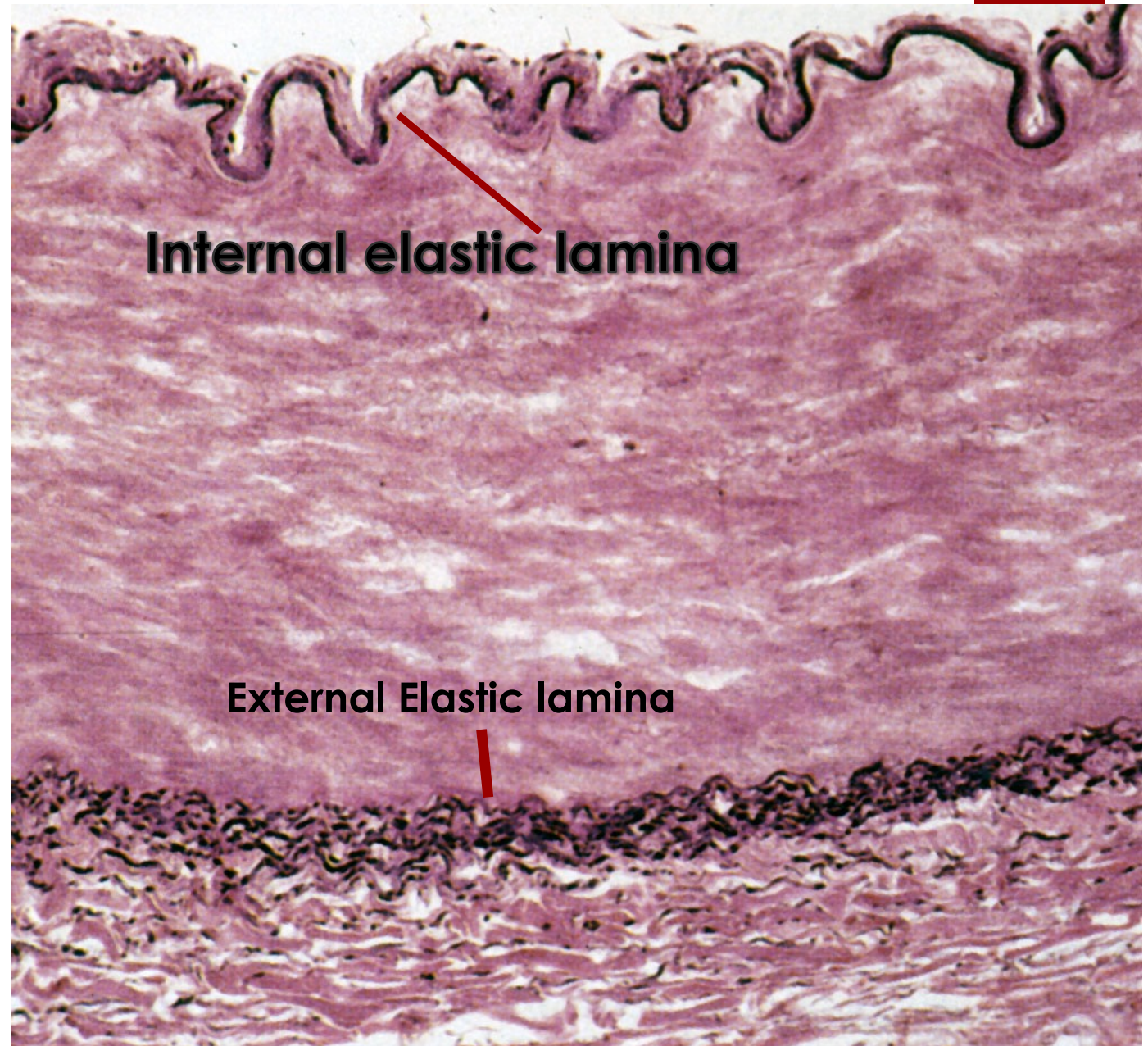
Tunica Media

Up to 40 layers of smooth muscle cells, thick external elastic lamina.

Tunica Adventitia

Thin layer of elastic connective tissue with sparse vasa vasorum

Muscular Artery



Arteriole – about 150 μm diameter and smaller – do not feel pulse. Blood is in continuous flow

Tunica Intima

Endothelium with basal lamina, minimal subendothelial layer, few elastic fibers without well-defined elastic lamina.

Tunica Media

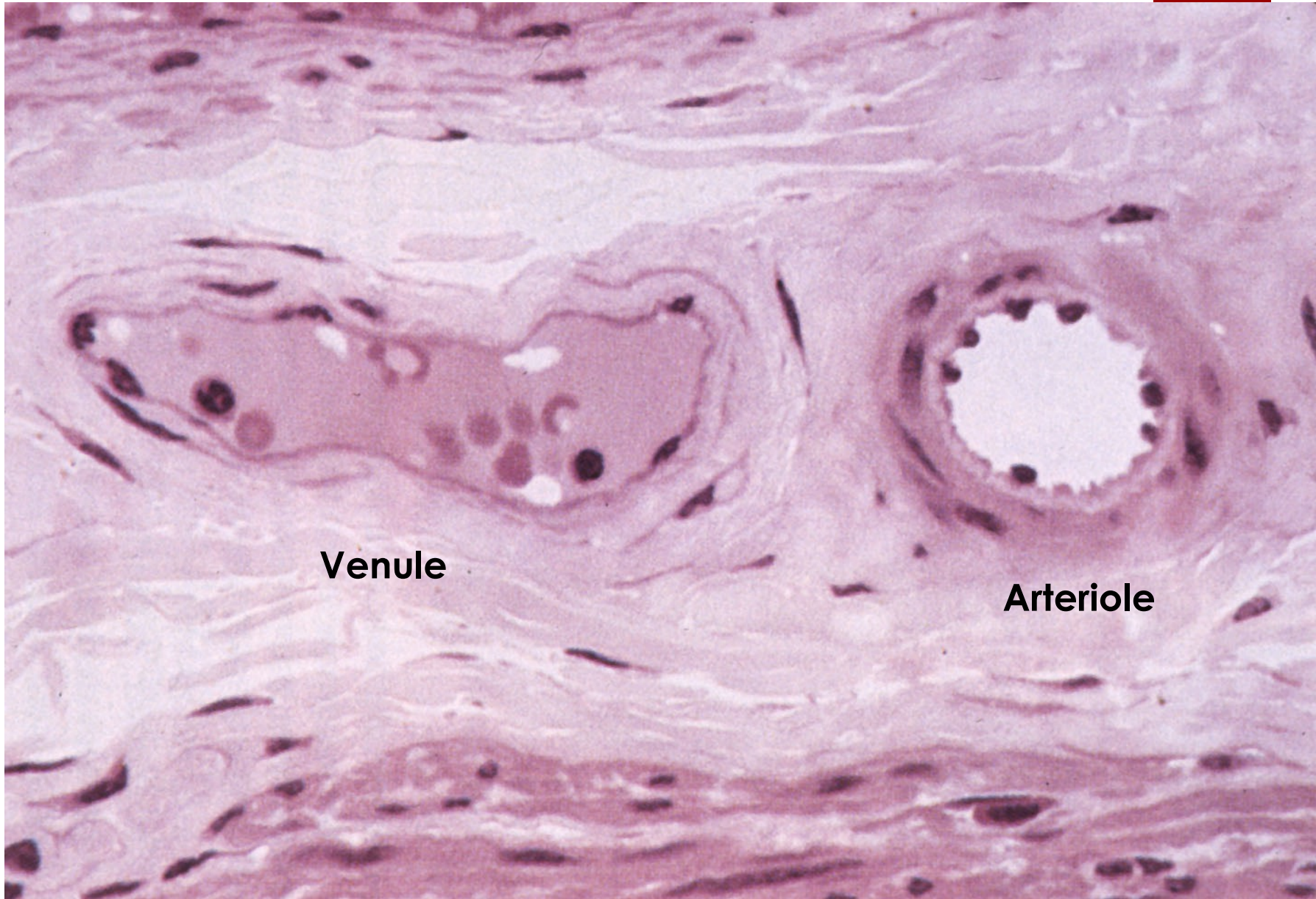
1-2 layers of smooth muscle cells

Tunica Adventitia

Loose connective tissue

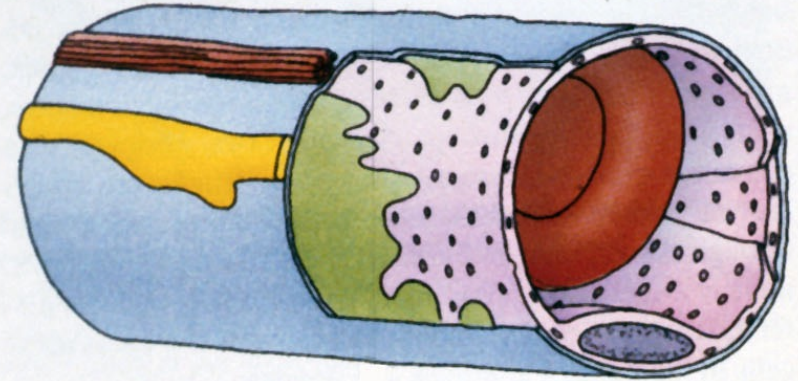
IMPORTANT: The posterior ciliary arteries are muscular arteries. The central retinal artery, after its first few mm, is an arteriole!!

Arteriole

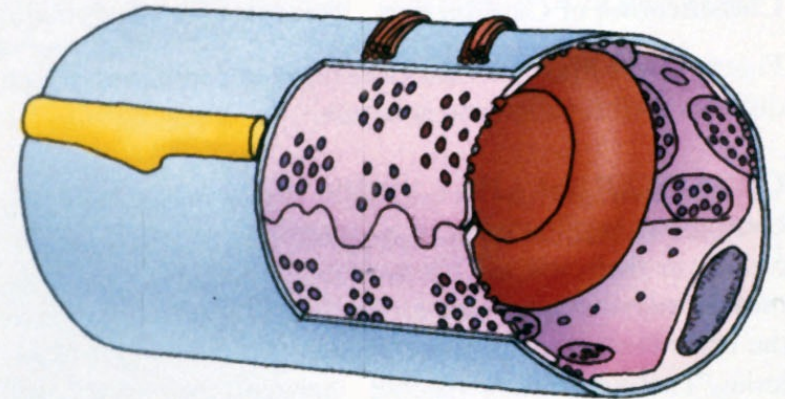


Capillary

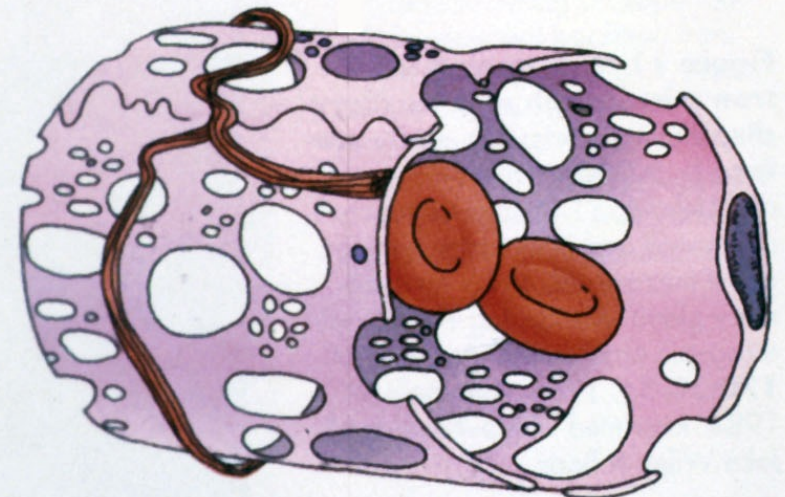
- Continuous (endothelial cells closed by tight junctions)
 - (e.g. iris, retina)
- Fenestrated (fenestra and leaky junctions (e.g. choroid, ciliary body))
- Discontinuous
 - (e.g. liver sinusoid, glomerulus)



A Continuous capillary



B Fenestrated capillary



C Sinusoidal (discontinuous) capillary

Venule

Tunica Intima

**Endothelium, basal lamina.
(pericytes) in post-capillary
venules**

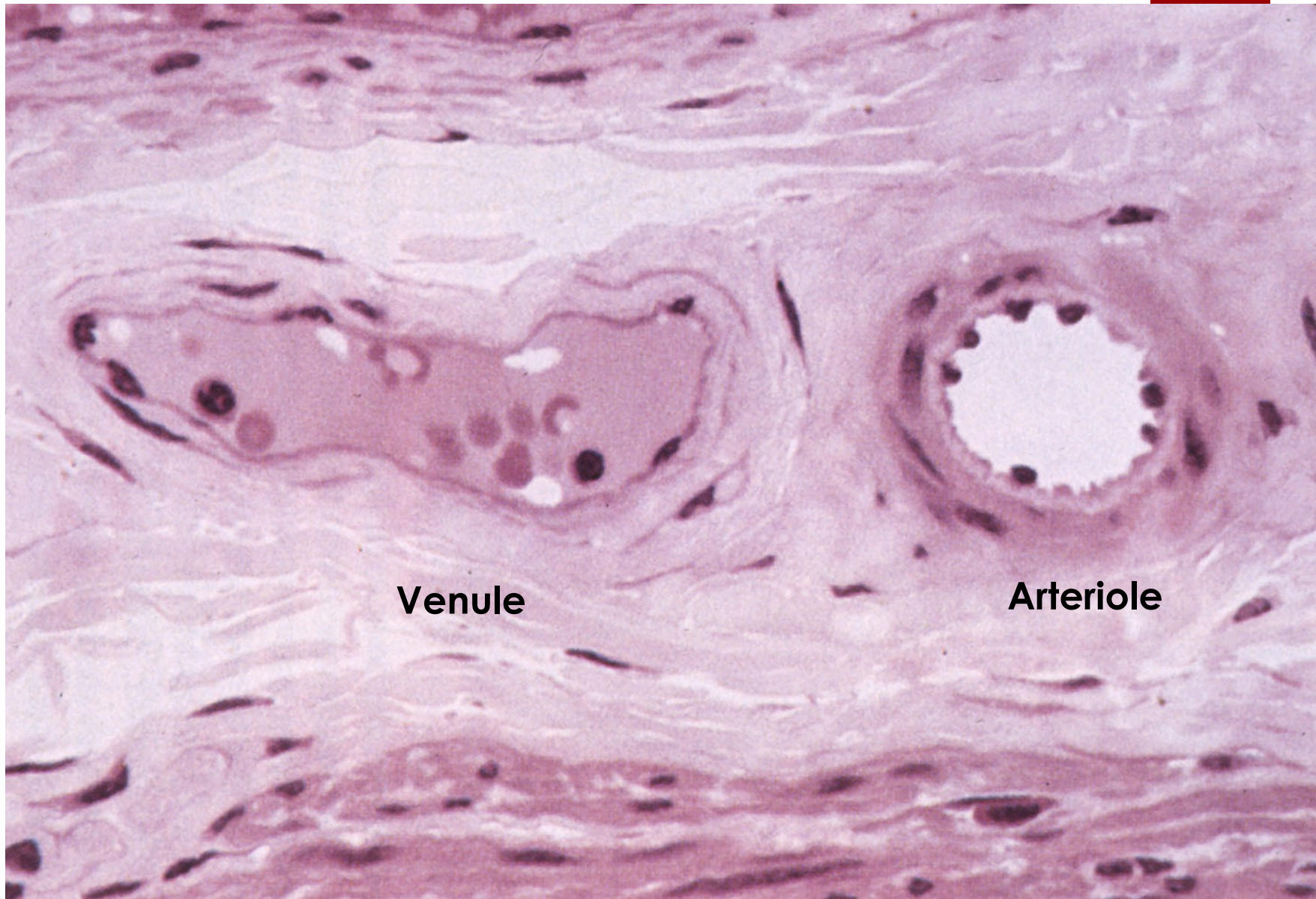
Tunica Media

**Sparse connective tissue and
scattered smooth muscle cells**

Tunica Adventitia

**Some collagen and scattered
fibroblasts**

Venule



Medium Vein – parallel with muscular arteries

Tunica Intima

Endothelium, basal lamina, valves below the heart, subendothelial connective tissue.

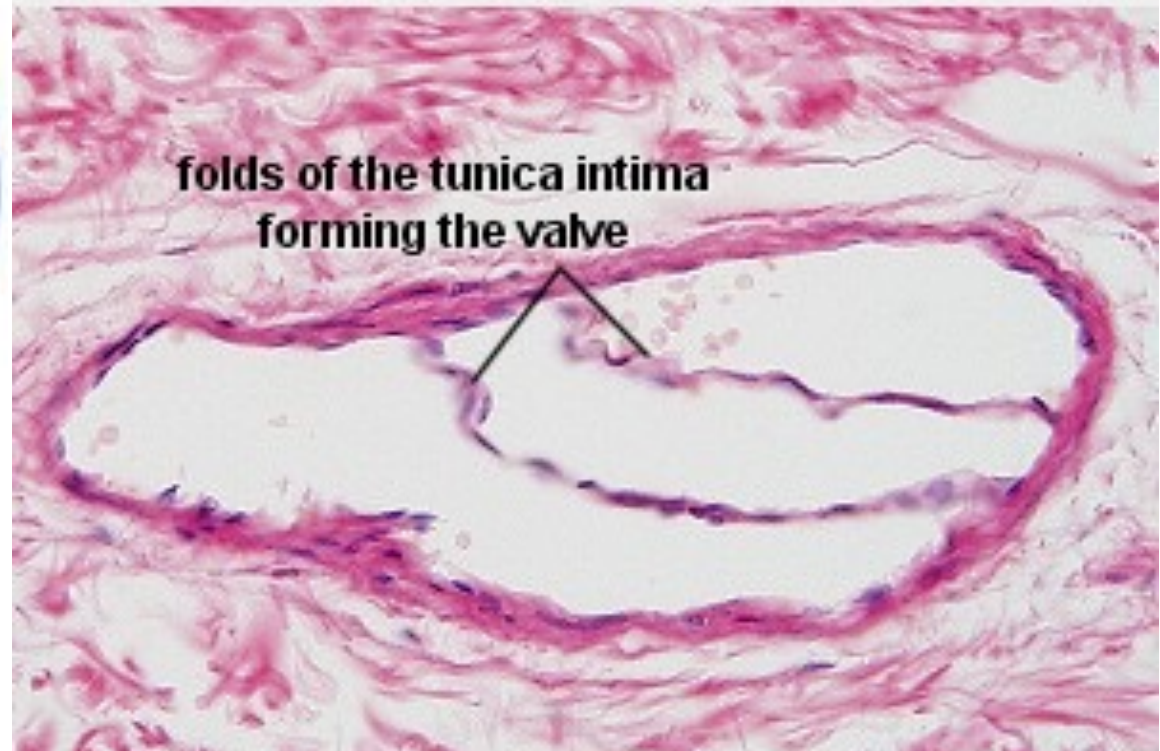
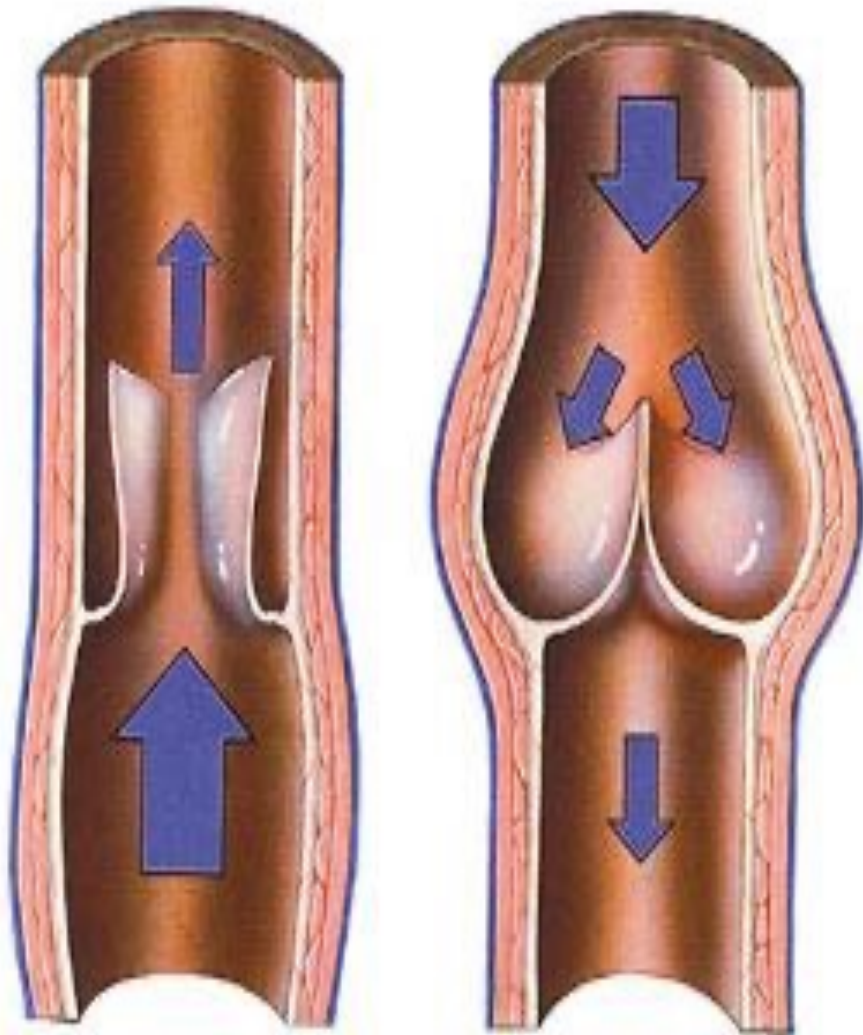
Tunica Media

Reticular and elastic fibers, some smooth muscle cells.

Tunica Adventitia

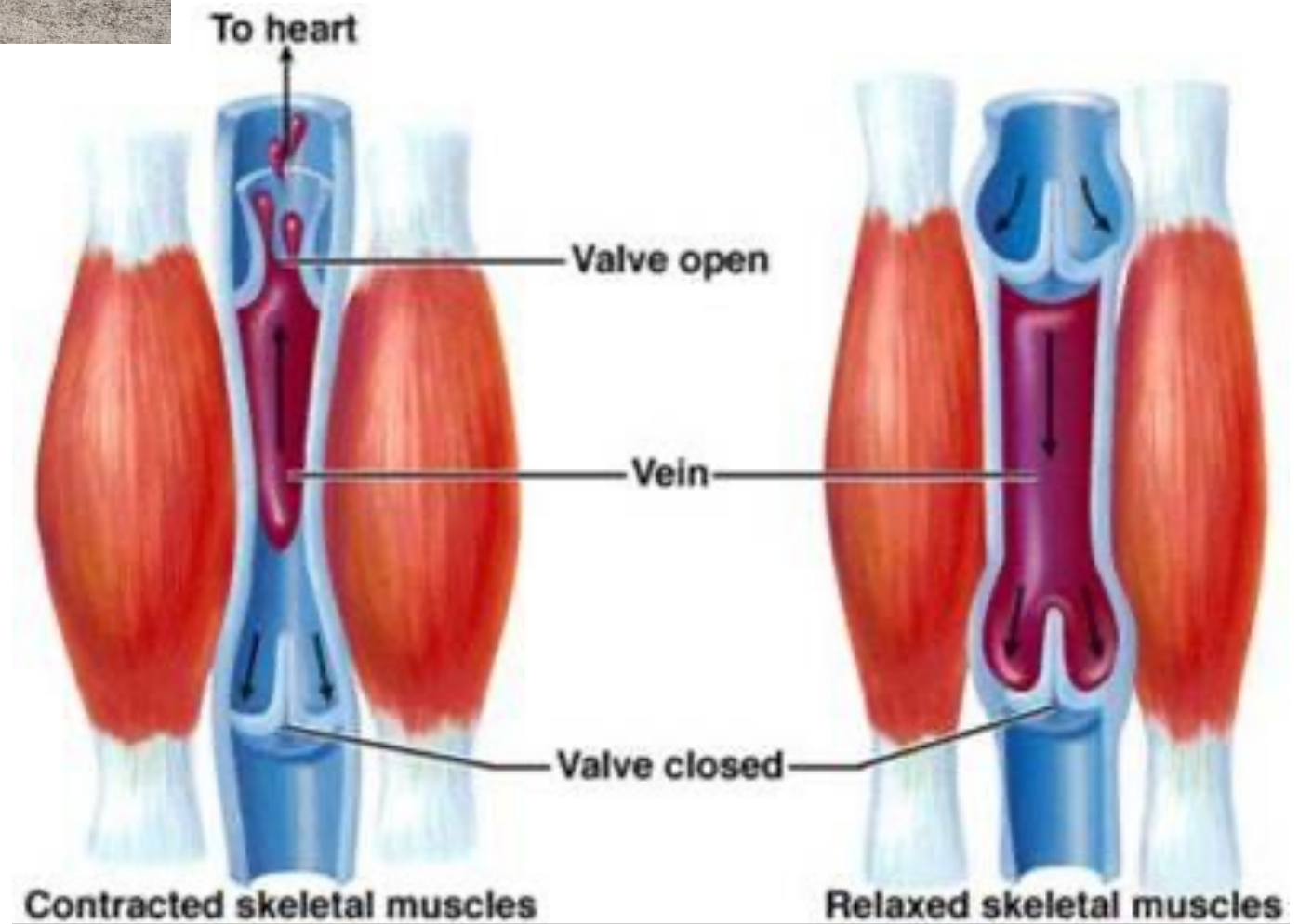
Collagen layers with fibroblasts

Veins Below but not Above the Heart

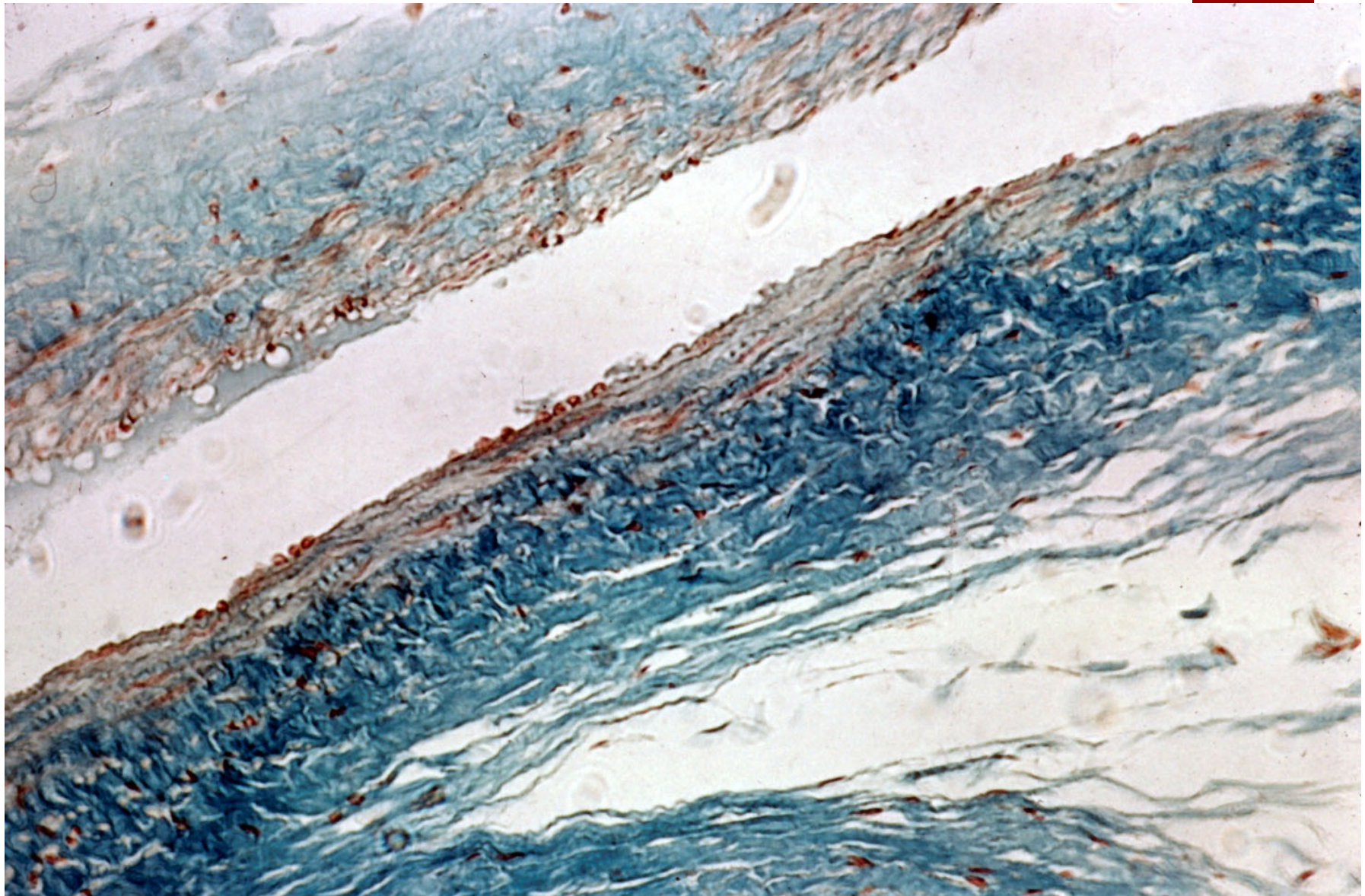




“Milking” of venous blood out of the legs. Soleus muscle in back of lower leg as “the second heart”



Medium Vein



Large Vein – parallel with Elastic arteries

Tunica Intima

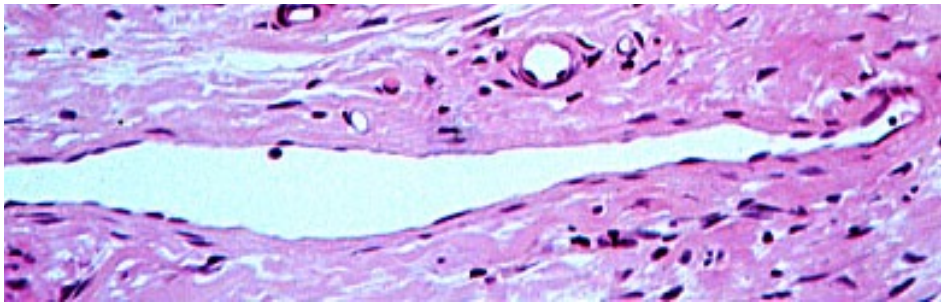
Endothelium, basal lamina, valves below heart, subendothelial connective tissue.

Tunica Media

Connective tissue, smooth muscle cells.

Tunica Adventitia

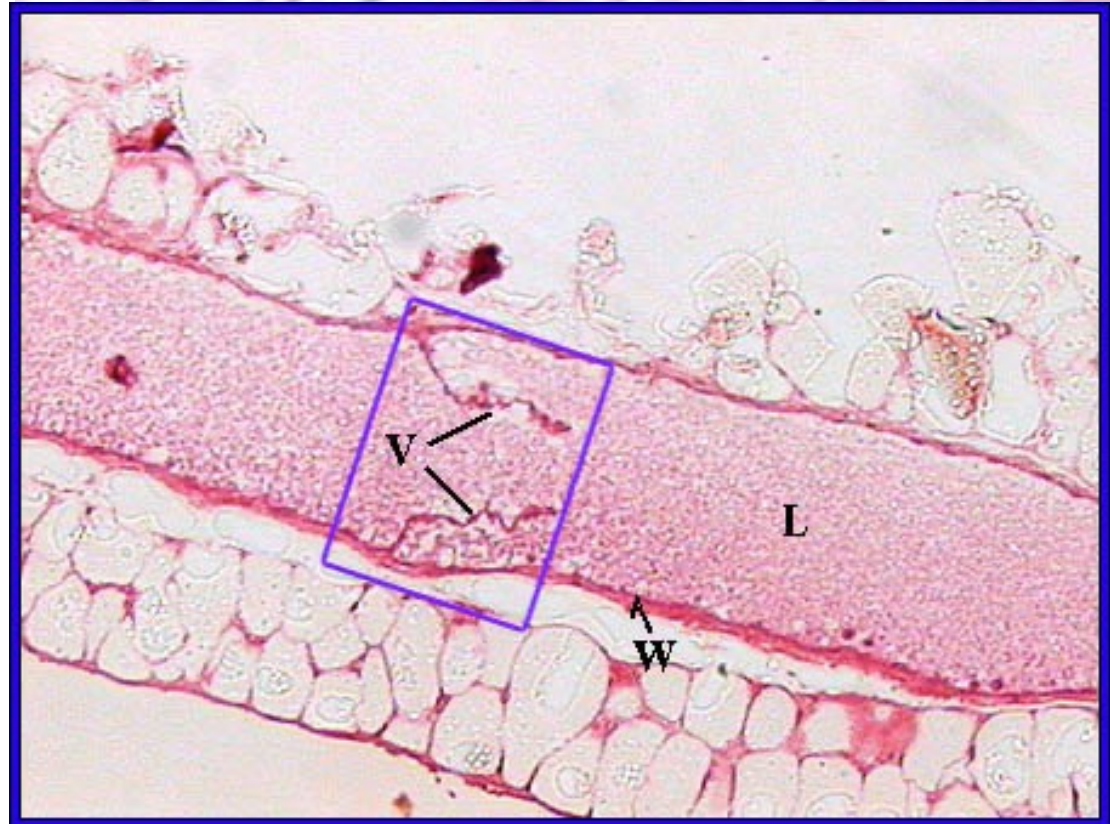
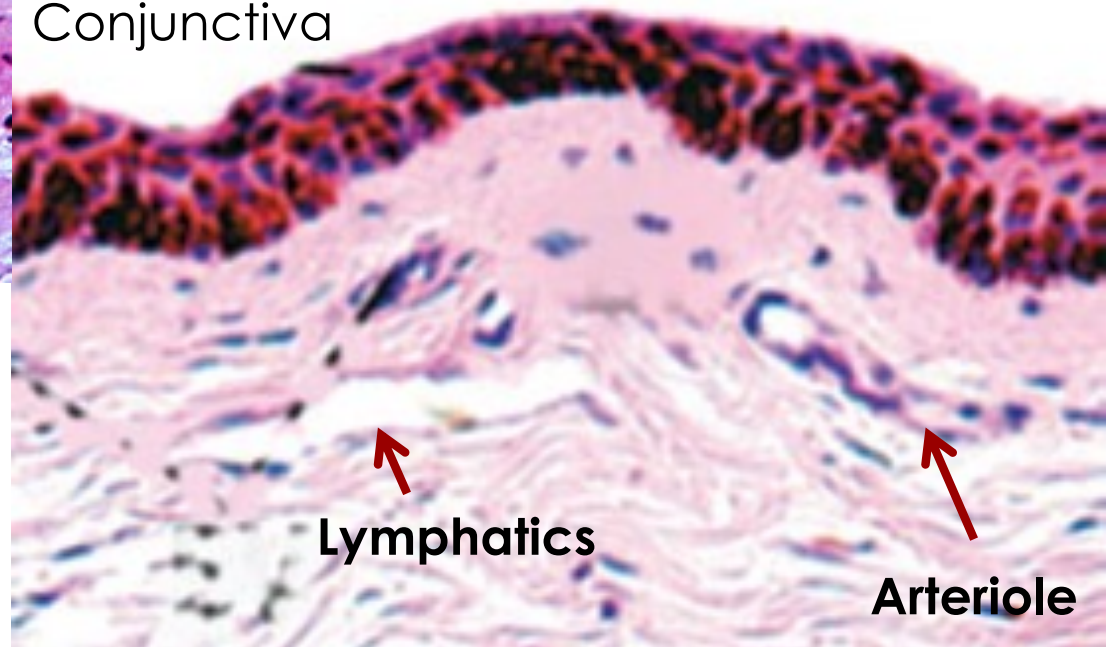
Smooth muscle cells oriented in longitudinal bundles



Lymphatics

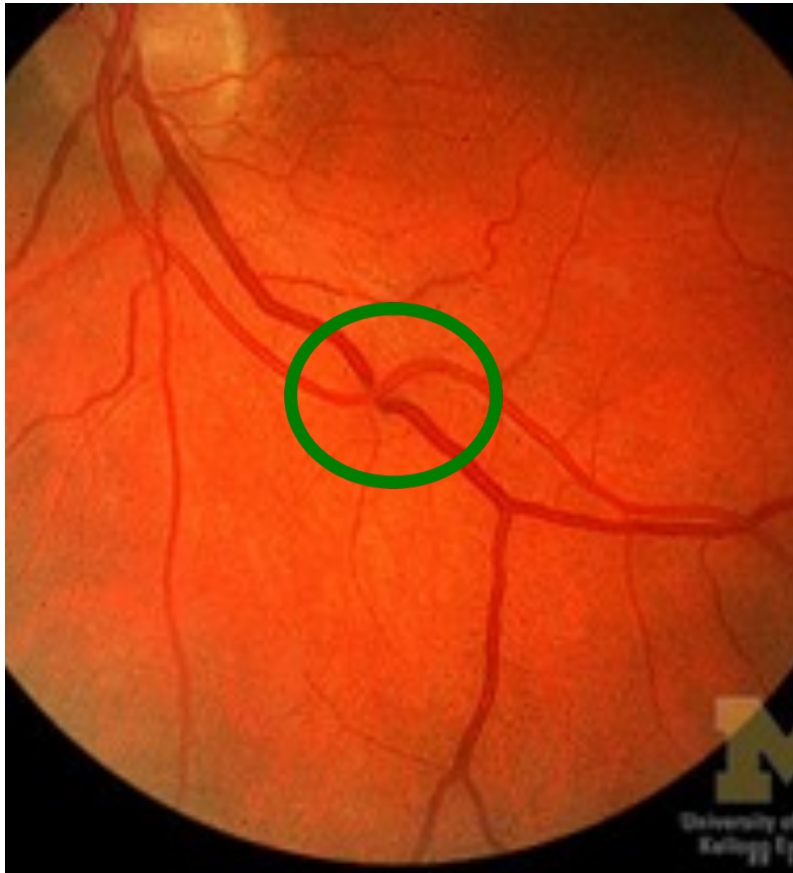
- Very thin walled lymphatic “capillaries” with tunica intima, minimal tunica media, more like a capillary.
- Larger Lymphatic channels have contractile elements in their media and they also exhibit valves.
- Endothelium has “button-like” junctions that allow fluid entry.

Conjunctiva

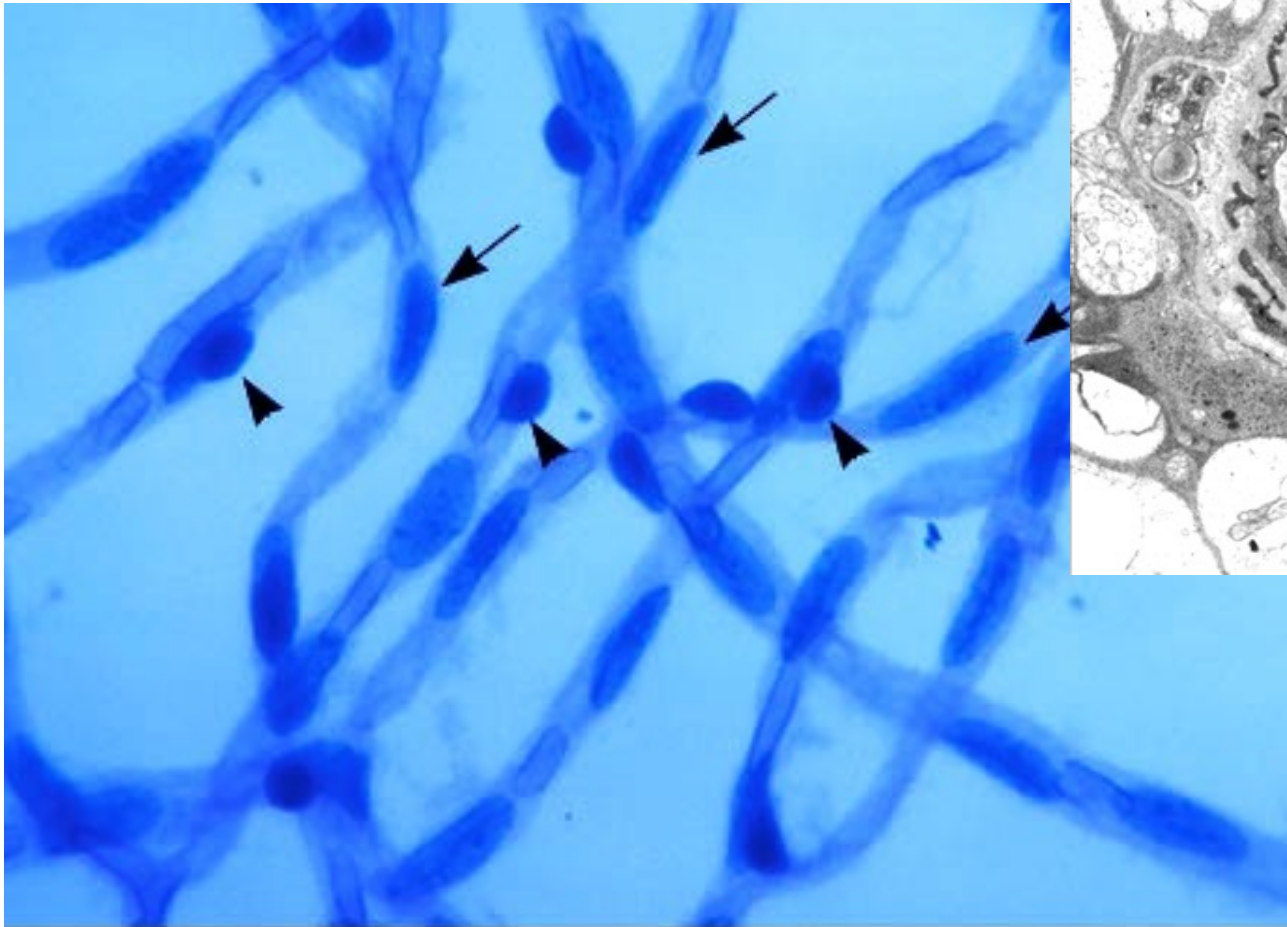


Arteriovenous Crossing in the Retina

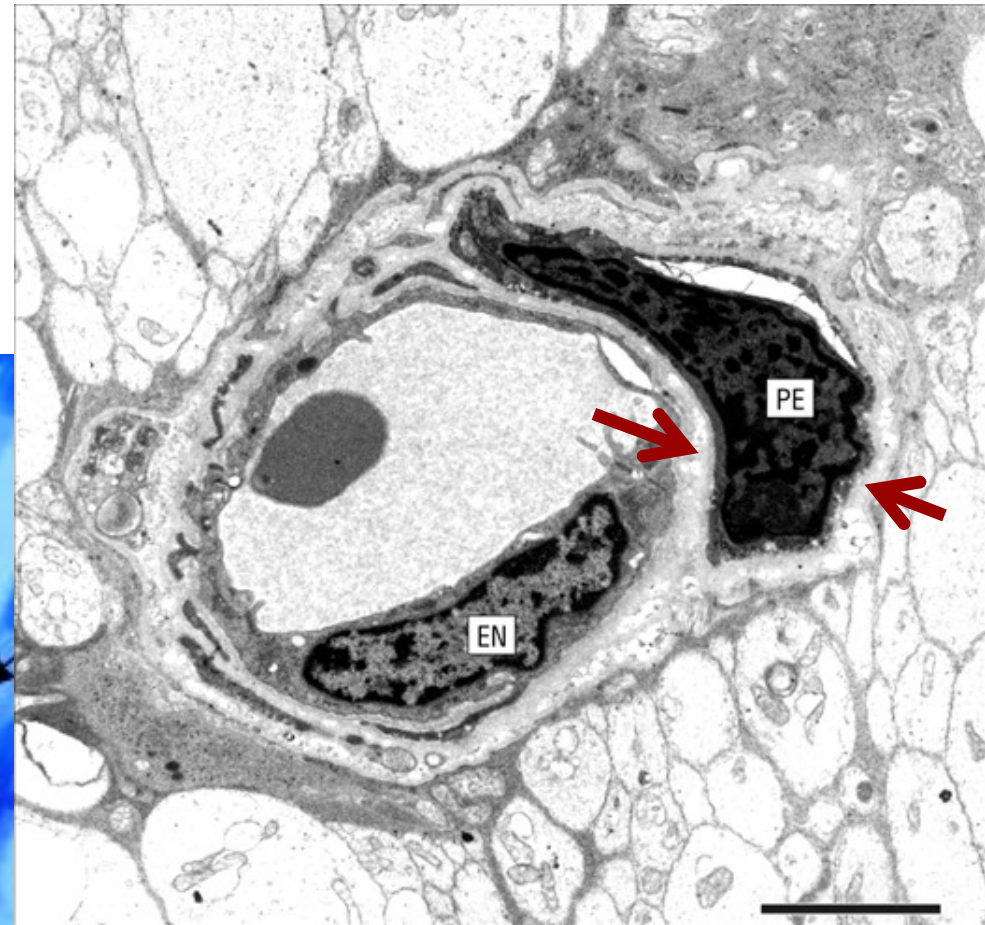
At arteriovenous crossings in the retina the two vessels share a common tunica adventitia. With advancing age, arteriolar sclerosis develops, producing thickening of arterial vessels. Since the venule cannot just slide out of the way, the vein gets focally constricted.



Retinal Capillaries

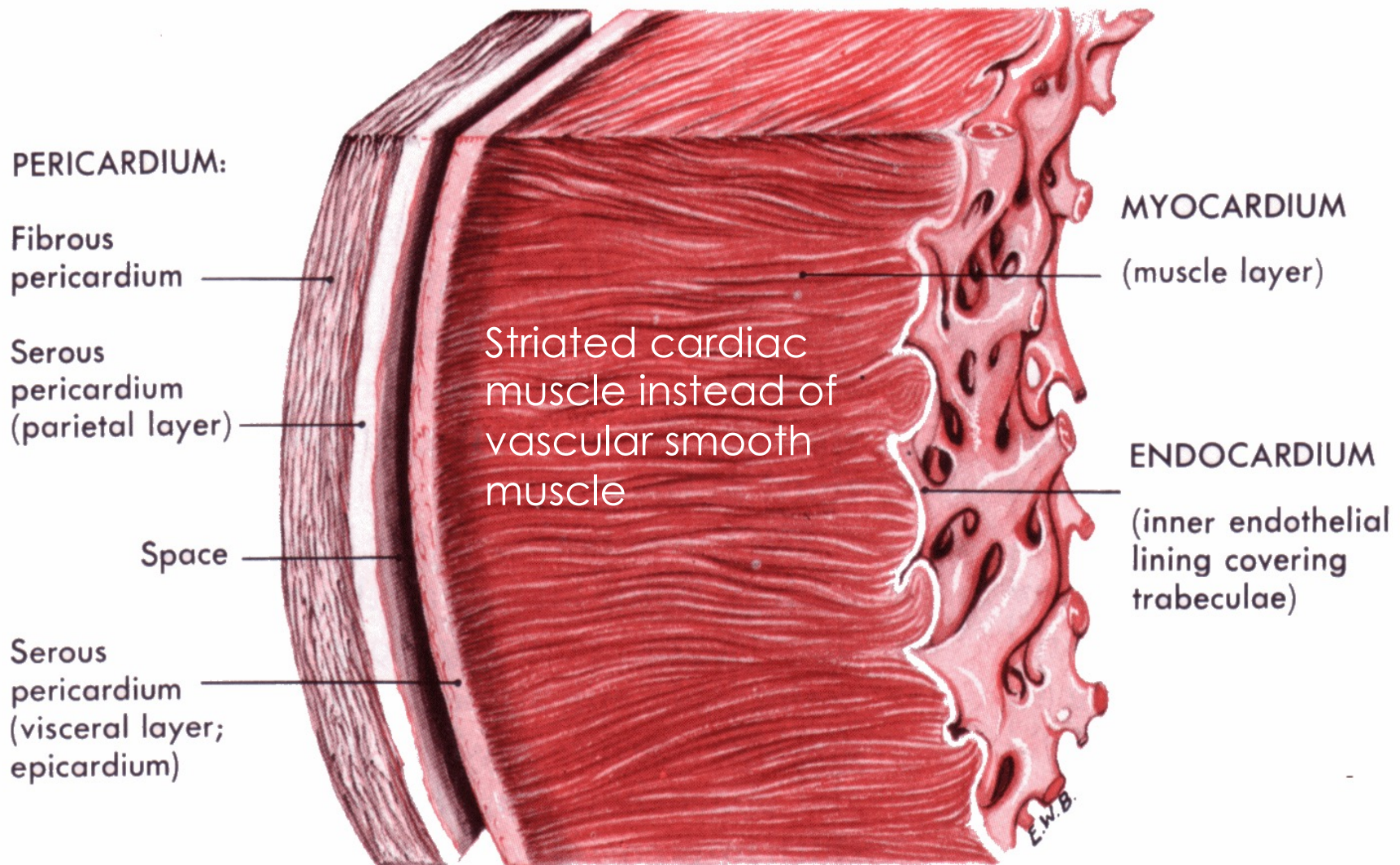


Normal retinal blood vessels, dog, PAS stain; trypsin digest; *arrowheads* - mural cells (pericytes), *arrows* - endothelial cells



Pericytes have some contractile properties. Unlike smooth muscle cells, pericytes are found within splits in the endothelial basement membrane. Smooth muscle cells are layered outside of the basement membrane.

Histology of Heart – analogy to tunica intima, media, adventitia



Section of the heart wall showing the components of the outer pericardium (heart sac), muscle layer (myocardium), and inner lining (endocardium).